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A Fourth Semester Project Proposal

On

CLOTHING MANAGEMENT SYSTEM

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ABSTRACT

Clothing Management System is a community-driven e-commerce marketplace designed to transform Nepal's digital fashion ecosystem by blending traditional retail with a circular economy focused on men's and women's apparel. The platform creates a vibrant, modern space where users can seamlessly buy, sell, and discover everything from trendy everyday menswear and womenswear to traditional cultural attire and affordable, pre-loved clothing. Built to tackle common frustrations in the local market, such as limited choices, outdated interfaces, and a lack of trust, the platform democratizes online retail by empowering independent creators, local boutiques, and ordinary individuals to clear out their closets or launch their own clothing brands. Key features include an instant buyer-to-seller toggle, virtual storefronts for local designers, smart filtering for clothing conditions and sizes, and secure escrow payments integrated with major Nepali digital wallets. The system also features end-to-end multi-vendor tracking and a multi-tier dashboard tailored for global admins, professional brands, and casual sellers. Developed a clean HTML/CSS/JavaScript frontend. It serves as a practical, learning-focused project aimed at making men's and women's fashion accessible, trustworthy, and uniquely Nepali.

Keywords: *Menswear, Womenswear, E-commerce Marketplace, Circular Economy, Multi-Vendor Platform, PHP/MySQl*

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1. Introduction

Clothing Management System is a full-fledged e-commerce web application built to disrupt the online fashion retailing experience in Nepal. The application aims to offer customers a broad array of trendy, affordable, and culturally inspired clothing, particularly for young adults and women who need both modern and traditional styles. In today's technology-oriented world, fashion retailing has become increasingly online; however, Nepalese people face various problems while shopping online, including limited product diversity in the local online shops, inefficient delivery mechanisms, bad UI/UX, absence of security while paying, and inadequate product information.

The Front end System runs perfectly well from the moment users log in to the platform and verify their account details. These credentials are verified with the central database for access authentication. Once the account details are successfully authenticated, buyers can perform a search for various clothing items and use filters to sort them according to size, condition, and price range. Buyers can then select items of interest to be added to their cart or saved to a wish list. At the payment step, the platform will either direct the user through the available digital wallets like eSewa and Khalti or through cash on delivery (COD). Right after the transaction is completed, the system will update the database and provide users with a receipt of purchase.

The process of backend processing is present and is made specifically for administrators and independent vendors. After that, the store manager and the clothing brand licensed to operate within the store can log into the secure administrative panel and see logs of orders, list apparel products, and change metrics of the products. The system checks for any ongoing transactions on the marketplace; when a transaction is finalized, the payment gateway is verified, the inventory of the vendor is updated, and a business invoice is created.

2. Problem Statement

Even after the development of various websites like Daraz, Hamro Bazaar, SastoDeal, SmartDoko, and more, there are many problems related to online fashion shopping in Nepal. The customers face the problem of limited selection of fashionable clothing, wrong description of the product, low-quality images, getting wrong sizes of clothes, and getting damaged products. Besides, delivery is late most of the time, especially in areas other than urban areas, and the return and refund process is too complicated and time taking. Apart from these, the

customers also complain about a weak customer support system and no confidence about the quality of the products on these websites.

Looking at the problems faced by the sellers in the current fashion websites of Nepal, the local fashion brands and fashion designers find it difficult to run their businesses on these websites. Some of the problems that they face include high commission fees, lack of freedom regarding branding, lack of an inventory management system, manual order management, and lack of analytical tools.

The problems found on the major websites like Daraz and Hamro Bazaar indicate a clear market gap. There is an urgent requirement of a system called a Clothing Management System where customers get a satisfactory online experience in shopping, and the sellers get good backend tools to manage their fashion business smoothly.

3. Objectives

The objective of the project is as follows: -

- To develop and design a web based **Clothing Management System** with dedicated dashboards for administrators, brands, and casual sellers to efficiently manage products, inventory, and orders.
- To build a user-friendly platform that supports seamless product listing, advanced filtering, secure payments, and smooth collaboration among sellers.

4. Scope and Limitations

The system focuses on providing users with a comprehensive clothing marketplace website for men's and women's apparel. The scope covers frontend development using HTML, CSS, and JavaScript, paired with a procedural or object oriented PHP backend and a MySQL relational database to support core e-commerce functionalities like product listings, shopping carts, wishlists, secure local payment gateway integration, multi-vendor order tracking, and administrative dashboards. The platform initially targets the Kathmandu Valley and major urban cities, with scalability architected for future nationwide expansion, and user testing conducted via structured sample data. However, the system has certain limitations due to time and resource constraints; it excludes advanced AI-driven features like virtual try-ons, and native mobile applications are left as a future extension. Furthermore, logistics and delivery tracking are restricted to internal simulations rather than live third-party API integration initial

deployment is confined to local or cloud based test servers rather than a full production scale, and background data collection relies primarily on surveys and secondary data collection methods.

5. Research Methodology

5.1 Methodology

Development follows a structured sequence of stages: understanding what is needed, checking that the plan is realistic, and designing how the system will be built. The project uses the Waterfall methodology, where each phase is completed before the next begins. This suits the project because requirements are clearly defined from the start, the scope is fixed, and work proceeds within a set academic timeline.

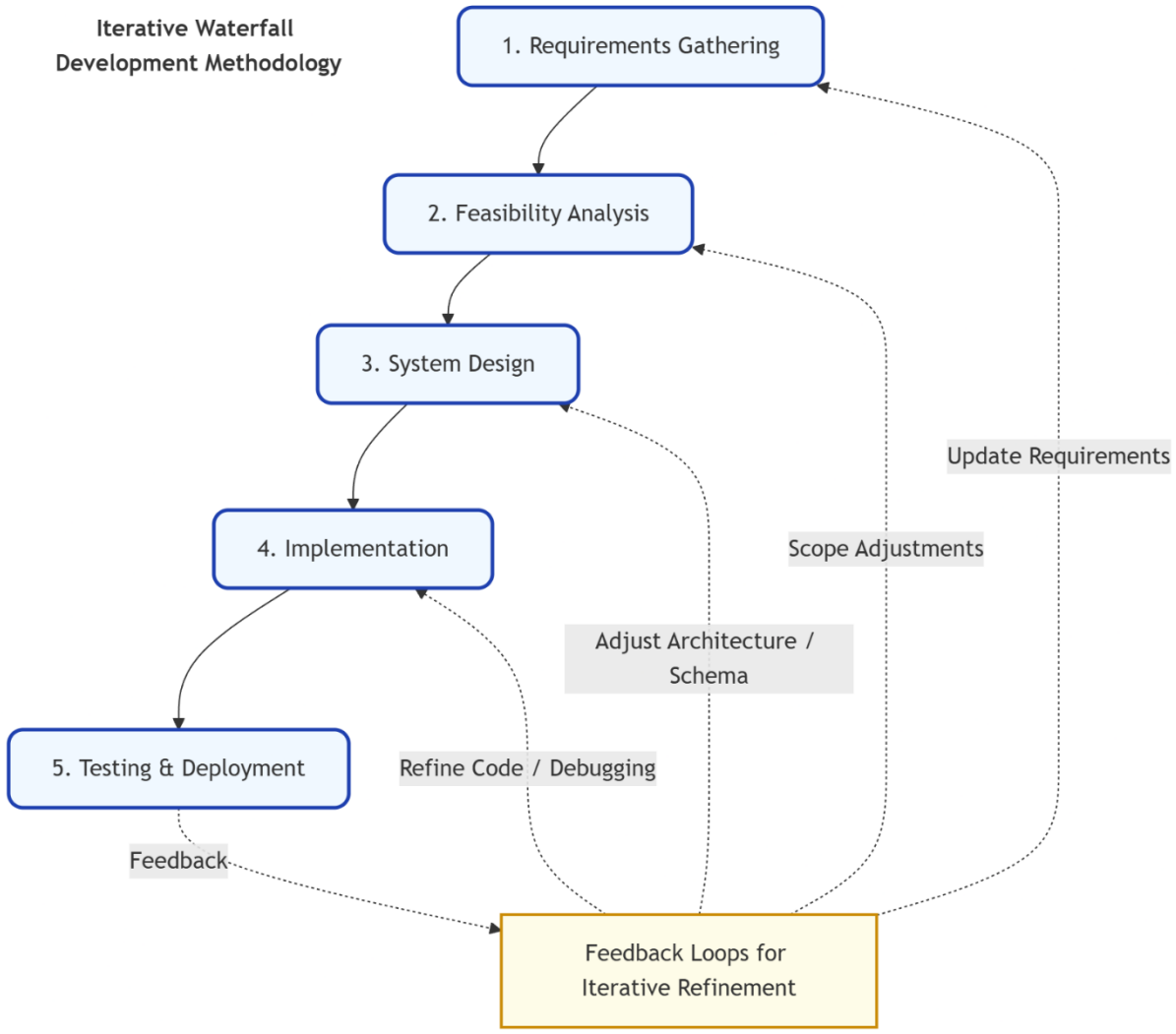


Figure 1: Waterfall Development Methodology (Clothing Management System)

- Requirements Gathering: This phase focuses on gathering and documenting the precise requirements of the desired capabilities of the e-commerce platform. This would involve defining user expectations, marketplace capabilities (such as a shopping cart and local wallet), and requirements for administrative management. If there are changes in requirements due to feedback during the development process, this phase is repeated for Updating Requirements.
- Feasibility analysis: At this stage, the project plans are checked for feasibility under academic constraints. Feasibility analysis includes technical feasibility, operational feasibility, economic feasibility, and schedule feasibility. If any problems in development process make the requirements too ambitious, the feedback loop allows for Scope Adjustment to bring the project back to feasibility.
- System design: This involves turning requirements into a technical design of the e-commerce platform. It includes developing the architectural design of the website, the layout of the user interface, and entity relationship diagrams for the database. If any backend database and/or structure problems arise during coding, the developer goes back to this phase for Adjusting Architecture / Schema.
- Implementation: This involves the actual coding process. The frontend is built using HTML, CSS, and JavaScript. Backend and multivendor operations are coded using PHP and MySQL. In case there are structural or logical bugs found by system testing, the process comes back to this phase for refining code and debugging.
- Testing & Deployment: This is the last phase where the built website undergoes testing to check for any security problems, dead links, or calculation errors using a structured sample dataset. The website will then be deployed in a local or cloud environment. Feedback from evaluators or users at this phase goes directly to the main Feedback Loop for iterative refinement.

5.2 Literature Review

Haque et al. investigated the structure and difficulties of operational e-business platforms in South Asian markets, paying attention to multi-vendor functions on Daraz. Catalog management, secure onboarding of vendors and order completion are identified as the basic components of a sustainable online marketplace. Based on these findings, this project provides separate backend routing panels for global administrators and vendors to manage their stock independently [1].

Dhakal and Dhungana studied the customer behavior and trustworthiness of online e-commerce platforms such as Daraz in Nepal. They argue that customer loyalty and adoption rate depends on reliable platform, local content filtering and transaction framework. In response to this, this project will apply condition-based filtering and size matrices to increase the clarity of products and build trust among local consumers [2].

Bajracharya conducted research on web application security from the perspective of server-side session management procedures in procedural and object-oriented web frameworks. He argues that reliable server-side session isolation is necessary for preventing privilege escalation. This project uses his recommendations, using PHP session variables for total isolation between administration module and standard privileges of vendors and customers [3].

Joshi researched techniques to optimize web applications from the server-side perspective, indicating the importance of secure validation scripts and input sanitization for protection against structural injections. Following these native PHP security standards, this project applies parameterized queries and form validation in order to securely manage multi-vendors' inventories and shopping cart functionality [4].

According to the documentation of eSewa Developers, manual payment verification systems create significant risks and operational costs for online stores. In this regard, this document describes automated status log of merchant's wallet, which verifies successful digital payment immediately. These findings were used in development of platform's checkout routing, using simulation of callback parameters to verify customer's transaction and update database accordingly [5].

Also, Khalti Developer Documentation gives standardized and secure transaction methods via merchant's payment gateways. It claims that availability of multiple checkout options greatly reduces transaction frictions among local consumers. Using this information, the payment

processing handler in the project was designed to switch between mock API payment parameters and Cash on Delivery states [6].

5.3 Feasibility Study

A comprehensive feasibility analysis was conducted across technical, economic, operational, and schedule dimensions to ensure the viability and successful execution of the project.

i. Technical Feasibility

The technical feasibility of this project can be attributed to its use of web technologies that are both fundamental and commonly available. The project makes use of HTML5 and CSS3 in terms of structuring and styling of the client side, together with JavaScript for client-side interaction. For the server side, PHP with a MySQL relational database management system has been used for implementation of the back end functionalities and forms processing. Considering that these are the commonly accepted standards in the field of web development, it means that there is abundant documentation and frameworks that are available.

ii. Operational Feasibility

The system is operationally feasible as it is designed with a simple and intuitive interface that requires no special technical training for users. Buyers can easily browse, filter clothes, and add items to their carts. Since the system features distinct panels, administrators can manage overall platform data, while local brands and casual sellers can easily add or update their own clothing listings without complexity. Furthermore, the system integrates familiar local digital payment gateways like eSewa and Khalti alongside Cash on Delivery, which matches the current online shopping habits of users in Nepal.

iii. Economic Feasibility

This project is highly cost-effective and economically viable since it relies entirely on free, open-source development tools and programming languages, including HTML, CSS, JavaScript, PHP, and MySQL. The development is carried out on existing personal computer hardware, eliminating any need for extra financial investment or infrastructure setup costs. For future sustainability, the platform can be hosted on affordable or free cloud servers

like Netlify for the frontend and standard shared hosting for the backend PHP environment.

iv. Schedule Feasibility

The project is operationally achievable within the allocated academic semester timeline. The entire development lifecycle is broken down into structured phases, including requirements analysis, database schema design, frontend coding, backend programming, and testing. Since advanced functionalities like AI try-ons or native mobile apps are excluded from the scope, the core multi-vendor e-commerce features can be comfortably completed, tested, and ready for defense on schedule.

5.4 Analysis and Design

Requirement Analysis for the clothing site is the process of defining the expectations of the users for an online apparel retail application that is to be built or modified. It involves all the tasks conducted to identify the shopping and administrative needs of different stakeholders, analyzing, documenting, validating, and managing these software requirements to facilitate system design. High-quality requirements for this e-commerce platform must be documented, actionable, measurable, testable, and traceable to capture digital business opportunities. The requirements for this platform are analyzed in two distinct ways: Functional Requirements and Non-functional Requirements.

a) Functional Requirement

The Clothing Website Online Store provides its users with the facility to sign up, log in to the system, and have control over their personal accounts, shipping information, and purchase history. The software lets the customers browse through an ever-changing product list, filter the clothes by size, color, or cost, control the products in the virtual shopping cart or wishlist, and complete the purchase process. In addition to that, the application offers secure online transactions, issues electronic invoices or receipts for the purchases, cancels or returns the purchased items, and sends automated notifications like shipment status and order confirmation messages. In terms of administration, the tool enables store owners to maintain details about clothing items, inventories, categories, and customers, and produce reports.

b) Non-functional Requirement

There is a security access that allows only certain people with specific roles to access sensitive information such as user details, payments, and system information. The system will help to synchronize the inventories in real time to avoid overselling of stocks. In addition, the site should be highly mobile responsive with maximum performance capability in order to handle peak traffic during promotions.

5.4 High-Level Design of System (Flow Chart)

A flowchart is a graphical representation of a process or system that uses symbols, shapes, and arrows to depict the sequence of steps or activities involved.

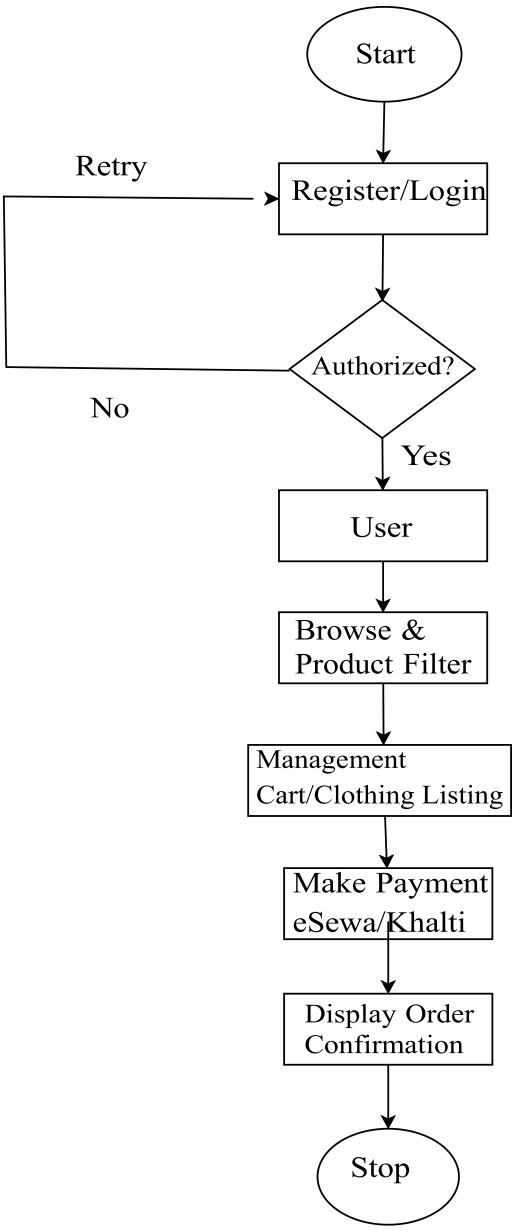


Figure 2: Flow Chart of Clothing Management System (User Section)

Flowchart Explanation of User Section[Paragraph]

- **Start**: The marketplace system initialization phase is triggered.
- **Register/Login**: The user provides authentication credentials or creates a new multi-tier account if not previously registered on the platform.
- **Authorization**: The system validates user login credentials against the database. If authorization fails, the workflow loops back to allow a retry; if valid, access is granted.
- **User Dashboard**: The user enters the localized account profile panel, where custom features are displayed dynamically based on account type.
- **Browsing and Filtering Products**: Users run search requests on the available inventory, filtering through clothing lines based on specific parameters and conditions.
- **Manage Cart / Clothing Listings**: Customers manage items within their active session cart while sellers modify active clothing listings and independent boutique inventory metrics.
- **Make Payment**: Transactions are securely routed using integrated digital wallets (eSewa or Khalti) or locked in as Cash on Delivery (COD).
- **Display Order Confirmation**: The system updates state logs inside the backend database and generates a dynamic digital summary receipt configuration for the user.
- **Stop**: System workflow lifecycle processes for the session safely conclude.

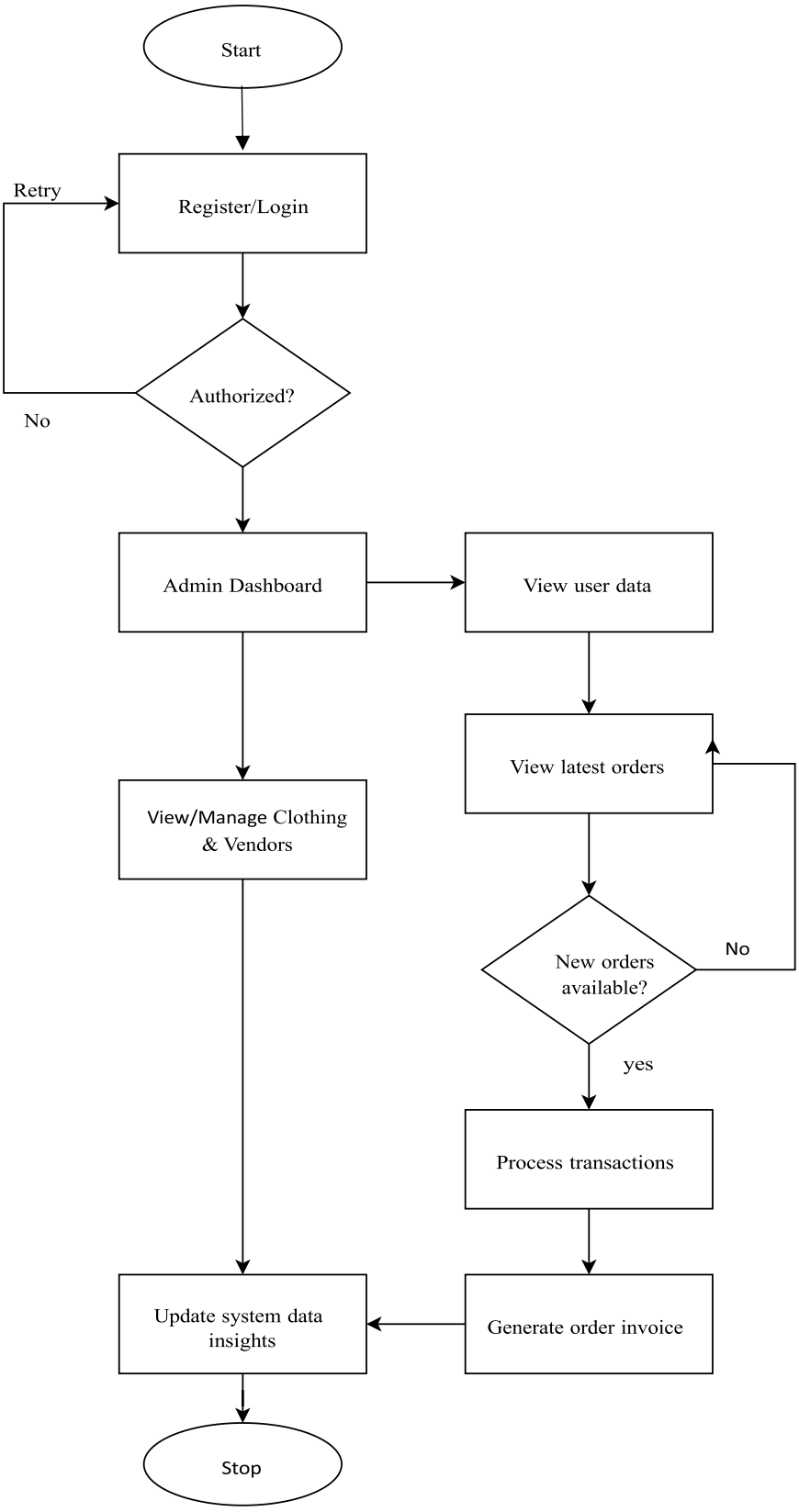


Figure 3: Flow Chart of Clothing Management System (Admin Section)

Flow Chart Explanation of the Admin Section

- **Start**: The admin system process is initialized.
- **Register/Login**: The administrator enters their credentials via the secure administrative portal interface.
- **Authorized?** The system validates the credentials against the database. If authorization fails, it loops back for a retry; if valid, access is granted.
- **Admin Dashboard**: The primary control hub loads, splitting management operations into parallel functional paths.
- **View/Manage Clothing & Vendors**: On the inventory branch, the admin views, adds, updates, or deletes vendor profiles and clothing product listings.
- **View User Data**: On the transaction branch, the admin views registered customer records and account details.
- **View Latest Orders**: The admin monitors the incoming order log to track purchase requests across the platform.
- **New Orders Available?** A decision check monitors for pending marketplace orders. If none are found, the system waits and loops back to check the log; if present, the process moves forward.
- **Process Transactions**: The system coordinates with integrated digital gateway logs (eSewa/Khalti) or logs Cash on Delivery statuses.
- **Generate Order Invoice**: A formal commercial digital invoice receipt is compiled and stored in the database.
- **Update System Data Insights**: Both operational branches converge to refresh analytics charts, vendor metrics, and the main dashboard sales statistics.
- **Stop**: The administrative process loop successfully concludes.

5.5 Implementation and Tools Used

The development of the platform utilizes a robust, industry-standard web stack chosen for its optimal relational efficiency, fast deployment capabilities, and minimal hosting overhead. The technical infrastructure relies heavily on server-side scripting and a structured data management framework to handle multi-vendor processes smoothly.

The specific software tools, programming languages, and environment components selected for implementing the system are detailed in the points below:

5.5.1 Development Tools & Technologies

- **HTML5:** Used for structuring pages semantically, designing layouts, and creating accessible forms.
- **CSS3 / Tailwind CSS:** Used for designing user interfaces that are modular and responsive to different mobile and desktop viewports.
- **JavaScript:** Used for doing client-side scripting, dynamic DOM manipulation, and asynchronous validations.
- **PHP (Hypertext Preprocessor):** Used for doing business logic processing, user session management, and routing of functional queries in the system.
- **MySQL:** RDBMS used for doing schema management and relational integrity constraints.

5.5.2 Environment & Infrastructure

- **XAMPP Environment:** Used for utilizing local Apache web servers and local database daemons during software configuration phases.
- **Apache Web Server:** Used for production environment infrastructure hosting to serve web scripts and handle public domain routing.
- **Local File System Storage:** Used for handling multi-vendor fashion asset management, secure product media uploads, and server-side directory pathing.

5.5.3 Design & Workflow Tools

- **Visual Studio Code:** Serves as the primary Integrated Development Environment (IDE) for clean codebase creation, local debugging, and syntax highlighting.
- **Draw.io:** Used for drafting high-level system designs, Entity-Relationship Diagrams (ERD), and formal operational Flowcharts.
- **Git & GitHub:** Used for local version control operations alongside cloud-backed staging repositories to ensure source code version histories.

6. GANTT Chart

Table 1: GANTT chart of Clothing Management System

Activities / Weekly	03/15	03/22	03/ 29	04/ 05	04/ 12	04/ 19	04/26	05/02	05/09	05/16	05/ 23	05/30
Initial Resource Check & Tech Setup	■											
Requirement Analysis & Research		■	■									
System Design & Relational DB Schema			■	■								
Frontend UI Layout Development					■	■						
Backend Programming & DB Entity Binding							■	■	■			
eSewa / Khalti API Sandbox Integration									■	■		
System Testing & Bug Refinement											■	
Comprehensive Report Documentation	■	■	■	■	■	■	■	■	■	■	■	■
Final Project Defense Preparation												■

7. Expected Outcome

This project involves creating an online marketplace for clothing among the Nepali community, where both new and used clothes can be bought and sold. The website will allow users to list their items and sort them by size and condition. There will be a multi-layer dashboard, which will give users the ability to change from buyer to seller and vice versa using the same account in the admin panel and a privately controlled seller store section.

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